JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Website: www.jdcoem.ac.in E-mail: info@jdcoem.ac.in
(An Autonomous Institute, with NAAC "A" Grade)
Affiliated to DBATU, RTMNU & MSBTE Mumbai
Department of Computer Science and Engineering
"A Place to Learn, A Chance to Grow"
Session: 2024-25

VISION	MISSION
To be recognized for excellent engineering, developing global leaders both in educational and research in the domain of computer science and wireless engineering.	1. To create self-learning environment by facilitating leadership qualities, team spirit and ethical responsibilities. 2. To improve department-industry collaboration, interaction with professional society through technical knowledge and internship program. 3. To promote research and development with current techniques through well qualified resources in the area of computer science and wireless engineering.

Question Bank

Semester/ Branch: III Sem/ CSE **Subject Code: CS3T001**

Subject Name: Computer Network **Subject In-charge: Prof. Gajanan Wankhade**

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Que. No	Questions	Unit	Marks
1	Define Data Communication.	**Unit-I**	2
2	Enlist the Application of computer network.	Unit-I	2
3	Identify the Components of Data Communication.	Unit-I	2
4	Define Data and Signal in computer network.	Unit-I	2
5	Restate the characteristics of Data Communication.	Unit-I	2
6	Define HUB in computer network.	Unit-I	2
7	State characteristics of computer network.	Unit-I	2
8	Identify Attenuation in the signal.	Unit-I	2
9	Define Topology with example.	Unit-I	2
10	State the Data Representation Methods in Computer Network.	Unit-I	2
11	Identify the types of Network.	Unit-I	2
12	Define Computer Network with its application.	Unit-I	2
13	Enlist Characteristics of Data Communication.	Unit-I	2
14	State the Data Representation Technique.	Unit-I	2
15	A given signal has frequency 256 HZ, 2000 HZ, 7000 HZ and 356 HZ determine the Bandwidth of Signal.	Unit-I	2
16	A given signal has Frequency 9000MHz, 4000MHz, 2000MHz and 8000MHz. Calculate the bandwidth of signal.	Unit-I	2
17	A signal has frequency 3125MHz, 254MHz, 9000MHz & 200 MHz Find the Bandwidth of signal.	Unit-I	2
18	Illustrate the need of Computer Network.	Unit-I	4
19	Describe the Component of Data Communication.	Unit-I	4
20	Rewrite Data & Signal with its types.	Unit-I	4
21	Explain Data flow with its types.	Unit-I	4
22	Discuss the types of Computer Network Architecture.	Unit-I	4
23	Write Network Hardware Component in details. .	Unit-I	4
24	Explain Topology with its different types in details.	Unit-I	7
25	Illustrate the types of Network in details.	Unit-I	7
26	Discuss Computer Networks with its Advantages and Disadvantages.	Unit-I	7
27	Illustrate the Network Hardware Component with different types of connecting devices.	Unit-I	7

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28	Explain the Topology with its different types in details.	Unit-I	7
29	Discuss the types of computer network.	Unit-I	7
30	Illustrate Data Communication with its Characteristics and Components in details.	Unit-I	7
31	Explain following types of Network. **a)** PAN **b)** LAN **c)** MAN **d)** WAN	Unit-I	12
32	Describe Computer Network with its types of Architecture and Advantages and Disadvantages of Computer Network in detail.	Unit-I	12
33	Differentiate between following: **a)** Data **Vs** Signal **b)** PAN **Vs** LAN **Vs** MAN **Vs** WAN **c)** Connectionless **Vs** Connection Oriented Service	Unit-I	12
34	Find the classes of following address: a) 11000001 10000011 00011011 11111111 b) 252.5.15.111	**Unit-II**	2
35	List out the transmission media in computer network.	Unit-II	2
36	Restate Propagation Delay.	Unit-II	2
37	Give example of Simple Parity Check.	Unit-II	2
38	Write short note on Transmission Impairment.	Unit-II	2
39	Define Data Rate Limit.	Unit-II	2
40	Enlist the types of Multiple Access Protocol.	Unit-II	2
41	State the services of Data Link Layer.	Unit-II	2
42	Find the Class of IP address 252.5.10.23.	Unit-II	2
43	Restate Definition of IP Address with example.	Unit-II	2
44	We need to send 165 Kbps over a Noiseless Channel with bandwidth of 20KHz. How many signal levels do we need?	Unit-II	4
45	We need to send 265Kbps over a noiseless channel with a bandwidth of 20Khz. How many Signal levels do we need?	Unit-II	4
46	Consider a Noisy Channel with a bandwidth of 3000Hz transmitting a signal with Two Signal Level. What can be maximum Bitrate?	Unit-II	4
47	Calculate weather 192.1.1.47 to 192.1.1.60 IP address is valid IP address block or not.	Unit-II	4
48	If we need to send 365 Kbps over a Noiseless Channel with a bandwidth of 30KHz. How many signal levels do we need?	Unit-II	4
49	Consider the following CRC generator and Find the corresponding binary pattern?	Unit-II	4

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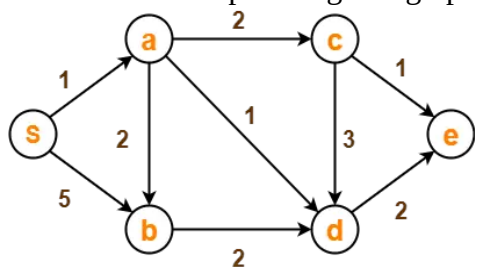
	a) $x^7 + x^6 + x^4 + x^3 + x + 1$ **b)** $x^7 + x^6 + x^3 + x^2 + x$ **c)** $x^5 + x^4 + x^2 + 1$ **d)** $x^{10} + x^7 + x^4 + x^3 + x + 1$		
50	Restate the Switching Techniques with its types.	Unit-II	4
51	Check whether 10.2.20.32 to 10.2.20.39 is Valid IP Address block or not?	Unit-II	4
52	Define Transmission Impairment with its types.	Unit-II	4
53	Write short note on Framing Method.	Unit-II	4
54	Discuss the Transmission Mode with its types in Computer Network.	Unit-II	4
55	Illustrate the Transmission Medium with its types in detail.	Unit-II	4
56	Discuss Error Detection Methods in computer network.	Unit-II	4
57	Find the Classes of Given IP address. **a)** 251.10.2.27 **b)** 20.10.1.10 **c)** 11111101 00011101 10101010 11100101 10101010 **d)** 10111000 11100011 00010101 10110001 11001010	Unit-II	4
58	Sketch Switching Techniques in computer network.	Unit-II	4
59	Discuss the Switching Technique with its types in detail.	Unit-II	7
60	Explain the types of error.	Unit-II	7
61	Examine the IP Address in computer network.	Unit-II	7
62	Consider the message sender wants to send is 100100, and the generator polynomial is $x^3 + x^2 + 1$. Find the message transmitted by the sender. If the receiver receives the message, check if the receiver receives the correct message or not.	Unit-II	7
63	Explain IP Address with types of Classful Internet Address in details.	Unit-II	7
64	Illustrate the Error Detection & Correction Methods in details.	Unit-II	7
65	Calculate weather following IP address is valid IP address block or not. a) 100.1.2.32 to 100.1.2.42 b) 192.1.1.47 to 192.1.1.60 c) 25.1.3.22 to 125.1.3.86	Unit-II	12
66	State the layer of TCP/IP Model.	**Unit-III**	2
67	Find out Shortest path of given graph from Source s to Destination e. 	Unit-III	2

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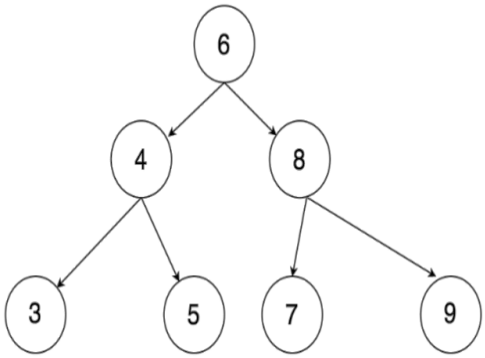
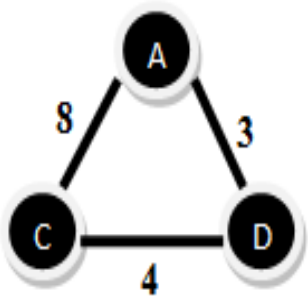
68	Rewrite Shortest Path Routing.	Unit-III	2
69	Define flooding in computer network.	Unit-III	2
70	Define Network Layer.	Unit-III	2
71	Enlist the Function of Network Layer.	Unit-III	2
72	State the types of Congestion Control Algorithm.	Unit-III	2
73	Identify protocols of Network Layer.	Unit-III	2
74	Find the Shortest Path of give tree. 	Unit-III	2
75	Illustrate the Network Layer Design Issues in details.	Unit-III	4
76	Write short note on Flooding Routing Algorithm.	Unit-III	4
77	Write short note on Congestion.	Unit-III	4
78	Find Distance Vector Routing of following routers weights. 	Unit-III	7
79	Describe the Routing Algorithms with its types in detail.	Unit-III	7
80	Consider the following network; three routers are there: A, B, and C Find the routing table for all the routers A, B, and C by using Distance Vector Routing.	Unit-III	7

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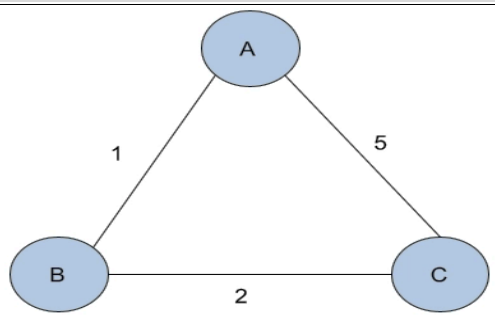
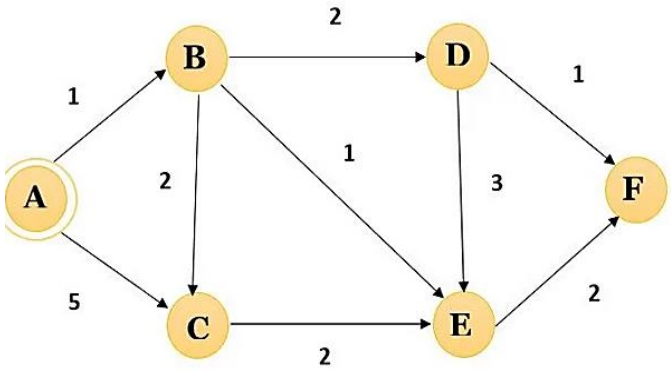
			
81	Describe Routing Algorithms with its types in details.	Unit-III	7
82	Restate the Services provided by Network Layer.	Unit-III	7
83	Distinguish between Connection Less Vs Connection Oriented Networks.	Unit-III	7
84	Determine the Shortest path of given graph from Source A to Destination F. 	Unit-III	7
85	Describe OSI Reference Model & TCP/IP Reference Model with its layers in details.	Unit-III	12
86	Describe the classification of Switching technique.	Unit-III	12
87	Explain OSI reference model in computer network.	Unit-III	12
88	Classify the Elements of Transport Protocols.	**Unit-IV**	2
89	Explain TCP Header with its Message Format.	Unit-IV	4
90	Discuss in detail TCP Timer with its types.	Unit-IV	4
91	List out the Elements of Transport Protocols.	Unit-IV	4
92	Write Sliding Window Protocol.	Unit-IV	4
93	State Functions of Presentation Layer.	Unit-IV	4
94	Rewrite types of TCP Timers.	Unit-IV	4
95	Distinguish between Go-Back-N ARQ and Selective Repeat ARQ.	Unit-IV	4
96	Discuss the Retransmission with its types in details.	Unit-IV	7
97	Write in detail the 3 way and 4 way of Handshake in TCP with diagram.	Unit-IV	7
98	Analyse TCP Header Format in Transport Layer Protocol.	Unit-IV	7
99	Explain with diagram Connection Establishment and Termination in TCP.	Unit-IV	7
100	List out the Application Layer Services.	**Unit-V**	2

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101	Write the Responsibilities of Presentation Layer.	Unit-V	2
102	Enlist Functions of Application Layer.	Unit-V	2
103	Classify the different Security Services in network.	Unit-V	2
104	Write short note on Digital Signature.	Unit-V	2
105	Examine the Architecture of WWW.	Unit-V	4
106	Write short note on WWW and HTTP.	Unit-V	4
107	Describe HTTP with different headers.	Unit-V	7
108	Explain Symmetric and Asymmetric Key Cryptography in detail.	Unit-V	7
109	Distinguish between following: **a)** Symmetric Key **Vs** Asymmetric Key **b)** Connection Less **Vs** Connection Oriented Networks **c)** TCP **Vs** OSI	Unit-V	12
110	Rewrite about following: **a)** SMTP **b)** TFTP **c)** DNS **d)** FTP	Unit-V	12
111	Explain the following layer:(any 4) **a)** Physical Layer. **b)** Data Link Layer **c)** Network Layer **d)** Transport Layer **e)** Session Layer **f)** Presentation Layer	Unit-V	12

Subject In charge **Dept. Academic Incharge** **Dept. Head AI**